

Family Name

Given Name

Student No.

Signature

THE UNIVERSITY OF NEW SOUTH WALES

School of Electrical Engineering & Telecommunications

FINAL EXAMINATION

SUMMER 2016 - 2017

ELEC1111 Electric Circuits

TIME ALLOWED:	3 hours
TOTAL MARKS:	100
TOTAL NUMBER OF QUESTIONS:	5

THIS EXAM CONTRIBUTES 60% TO THE TOTAL COURSE ASSESSMENT

Reading Time: 5 minutes.

This paper contains 6 pages.

Candidates must **ATTEMPT ALL** questions.

Answer each question in a **separate answer booklet**.

Marks for each question are indicated beside the question.

This paper **MAY NOT** be retained by the candidate.

Print your name, student ID and question number on the front page of each answer book.

Authorised examination materials:

Candidates should use their own UNSW-approved electronic calculators.

This is a closed book examination.

Assumptions made in answering the questions should be stated explicitly.

All answers must be written in ink. Except where they are expressly required, pencils **may only be used** for drawing, sketching or graphical work.

QUESTION 1 [20 marks]

- (i) Calculate the equivalent impedance of the circuit of Figure 1 as seen by :
 a. **(4 marks)** a *dc* voltage source
 b. **(6 marks)** an *ac* voltage source with a frequency of $\omega = 1$ rad/sec

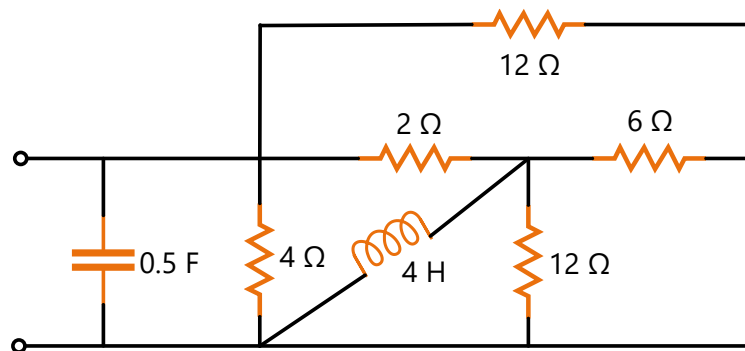


Figure 1

- (ii) **(6 marks)** Find the Thevenin equivalent of the circuit in Figure 2.

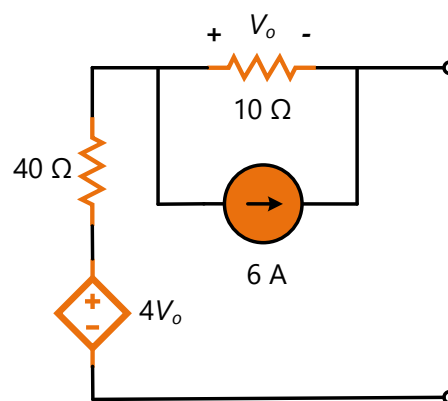


Figure 2

- (iii) **(4 marks)** Calculate the power of the three sources in the circuit of Figure 3. Clearly explain if a source supplies or absorbs power.

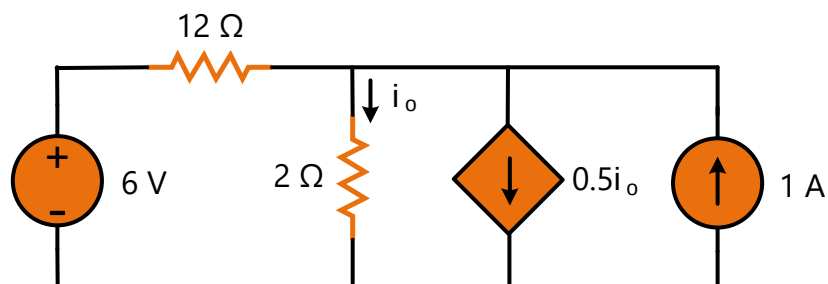


Figure 3

QUESTION 2 [20 marks]

- (i) **(6 marks)** Find the voltage across the $1\ \Omega$ resistor and the power dissipated in the $3\ \Omega$ resistor of the circuit in Figure 4.

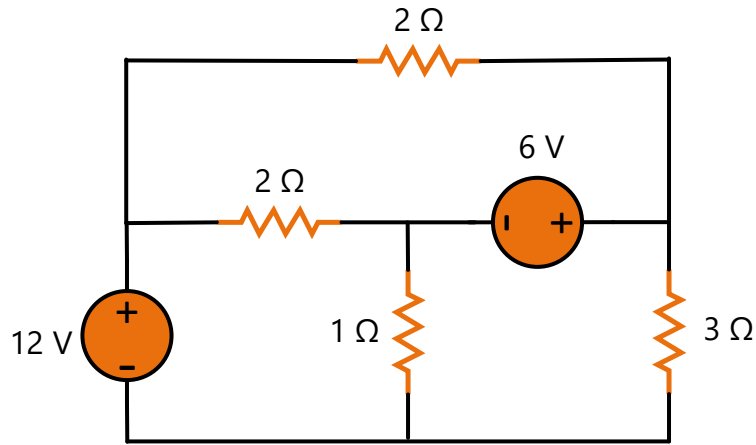


Figure 4

- (ii) **(5 marks)** Find the voltage of node x (V_x) in the circuit of Figure 5.

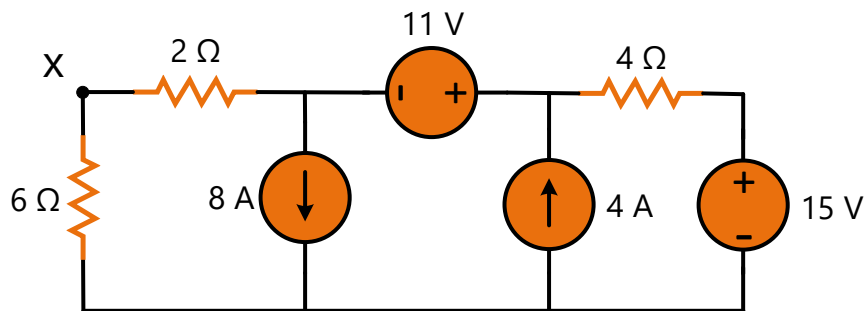


Figure 5

- (iii) **(9 marks)** Find the current $i_o(t)$ in the circuit of Figure 6.

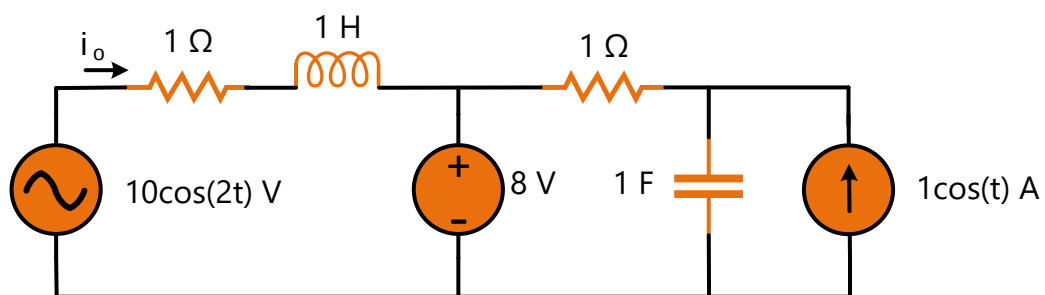


Figure 6

QUESTION 3 [20 marks]

- (i) **(8 marks)** In the circuit of Figure 7, the switch has been in the open position for a long time and closes at $t = 0$. Showing all working, derive an expression for the voltage of the capacitor $v_c(t)$ for $t > 0$.

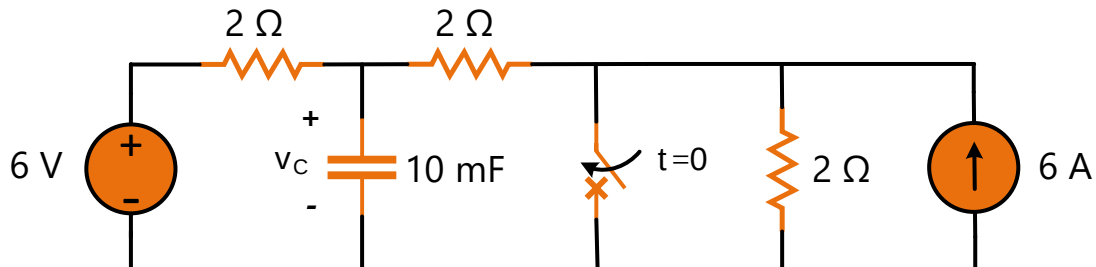


Figure 7

- (ii) **(8 marks)** In the circuit of Figure 8, the switch has been in the closed position for a long time. The switch opens at $t=0$ and closes again at $t=1\text{sec}$. Showing all working, derive an expression for the current through the inductor $i_L(t)$ for $t > 0$.

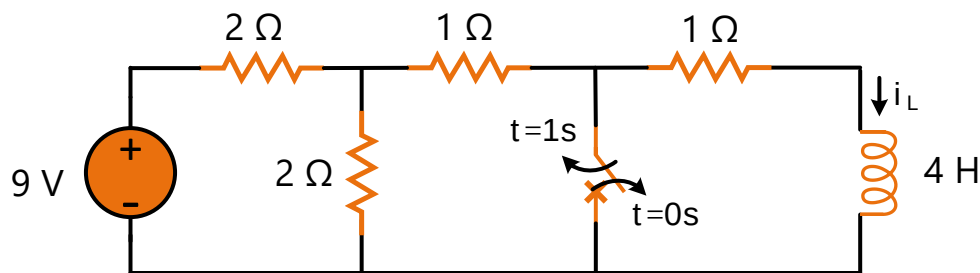


Figure 8

- (iii) **(4 marks)** Find the element and the value of the element that must be connected parallel to nodes a-b of the circuit shown in Figure 9 so that the impedance Z_{eq} , seen from an ac voltage source with a frequency $\omega = 10 \text{ rad/sec}$ is purely resistive.

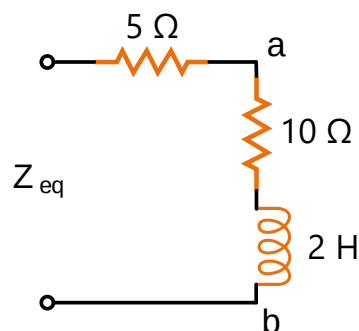


Figure 9

QUESTION 4 [20 marks]

- (i) **(5 marks)** Find V_o in the circuit of Figure 10 and draw phasors V_o , V_1 , and V_2 in the same phasor diagram.

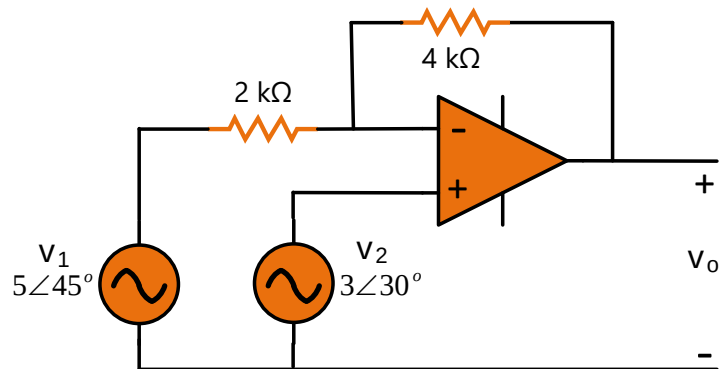


Figure 10

- (ii) **(7 marks)** Consider the circuit of Figure 11. Find the current I_o and the voltages of the resistor V_R , the capacitor V_C and the inductor V_L . Plot them in the same phasor diagram, clearly indicating each phasor.

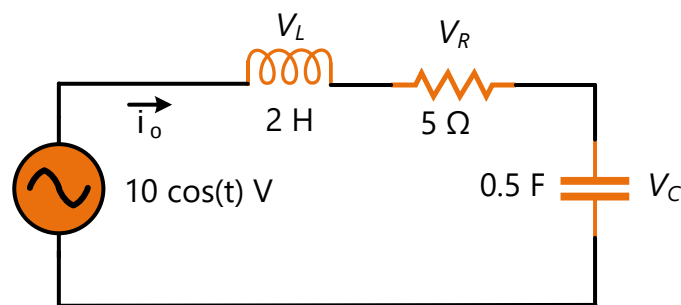


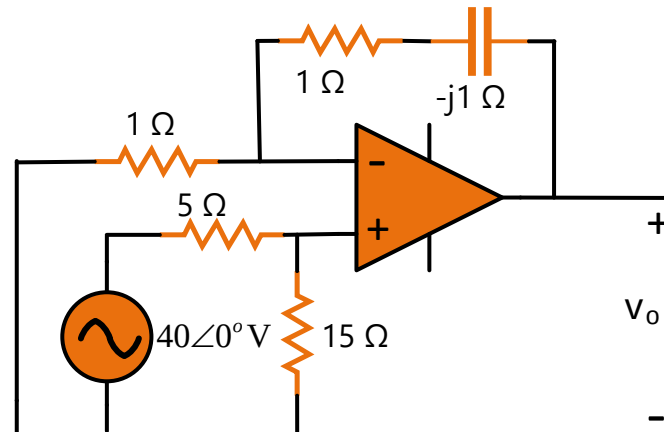
Figure 11

IGNORE THIS QUESTION

- (iii) **(8 marks)** Two loads are connected in parallel and are supplied by a $230 V_{rms}$, 50 Hz voltage source. **Load 1** consumes 10 kVA of power at a power factor of 0.8 leading (capacitive) while **load 2** consumes 5 kW at a power factor of 0.9 lagging (inductive). Calculate:
- The total complex power of the two loads.
 - The total apparent power.
 - The combined power factor of the two loads, and
 - The rms value of the current supplied by the source.

QUESTION 5 [20 marks]

- (i) **(5 marks)** Design an operational amplifier circuit where $v_o(t) = -15 v_s(t)$. Explain analytically your design process.
- (ii) **(7 marks)** For the circuit shown in Figure 12, calculate the output voltage V_o .



IGNORE THIS QUESTION *Figure 12*

- (iii) **(8 marks)** Consider the following logical diagram
- Derive a logical expression for Z.
 - Write the truth table of the circuit.
 - Explain, also showing the required connections and circuits, how a NAND Gate can be used to configure i) a NOT gate, ii) an AND gate and iii) an OR gate.

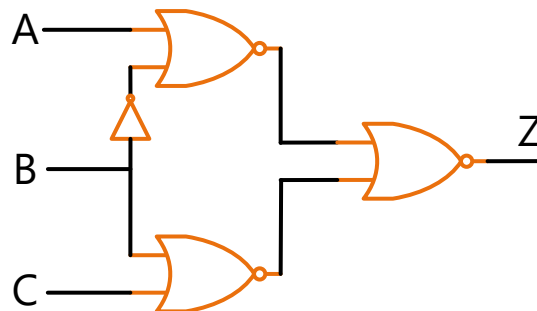


Figure 13

END OF PAPER